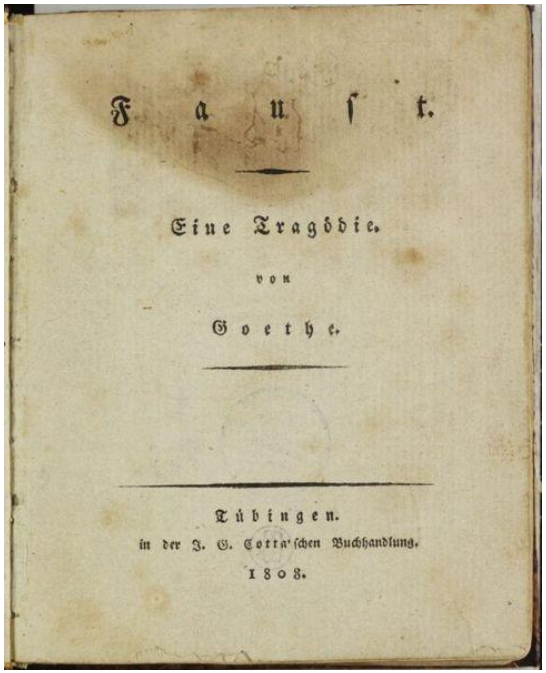
Chapter 2
Chemical Bonding

“Ach wüßt ich nur was die Welt, im innersten zusammenhält.”

- **Dr. Faust desperately seeking “…understanding of the true forces that are holding matter together on this earth” before the devil appears as Mephisto in *Johan Wolfgang von Goethe’s* “Faust” Part I**



Faust.

Eine Tragödie.

von

Goethe.

Tübingen.

in der J. G. Cotta'schen Buchhandlung.

1808.

Dr.Faus and Mephisto

Chemical Bonding

- Interaction between atoms leads to a rearrangement of the electrons to a more stable state - bonding.

- The types (extremes) of chemical bonds:

  - Ionic bond - electrostatic forces hold together oppositely charged ions.

  - Covalent bond - two atoms sharing electron pair(s). Electrons are mutually attracted by the nuclei of adjacent atoms.

  - Metallic bond -is a complicated type of bond. The elements are envisaged of being metal cations in a “sea” of electrons. The electrons are shared between all the ions simultaneously… (Not the whole picture!)

To understand chemical bonding, trends in three atomic properties must be considered:

- Ionization energy
- Atomic radius
- Electron affinity

Ionization Energy

Ionization energy is the energy required to strip an electron from a gaseous atom.

$$Mg_{(g)} \longrightarrow Mg^{+}_{(g)} + e^{-} \qquad I_1 = 738 \text{ kJ/mol}$$

$$Mg^{+}_{(g)} \longrightarrow Mg^{2+}_{(g)} + e^{-} \qquad I_2 = 1451 \text{ kJ/mol}$$

	Atomic Radius, pm	Ionization Energy (I_1), kJ/mol
Li	152	520.2
Na	186	495.8
K	227	418.8
Rb	248	403.0
Cs	265	375.7

Petrucci, Table 9.3 (pg. 253)

Ionization energy decrease as atomic radii increase.

Petrucci Table 9.4, p. 354

TABLE 10.4 Ionization Energies of the Third-Period Elements (in kJ/mol)

	Na	Mg	Al	Si	P	S	Cl	Ar
I_1	495.8	737.7	577.6	786.5	1012	999.6	1251.1	1520.5
I_2	4562	1451	1817	1577	1903	2251	2297	2666
I_3		7733	2745	3232	2912	3361	3822	3931
I_4			11580	4356	4957	4564	5158	5771
I_5				16090	6274	7013	6542	7238
I_6					21270	8496	9362	8781
I_7						27110	11020	12000

Electron Affinity

Electron affinity is a measure of the energy change that occurs when a gaseous atom gains an electron.

$$F_{(g)} + e^- \rightarrow F^-_{(g)} \qquad EA = -328 \text{ kJ/mol}$$

This reaction is exothermic.

<u>Trends</u>: generally, electron affinities are larger (more negative) towards the right of the periodic table and less negative towards the bottom of a group (some exceptions).

Sizes of Atoms and Ions

- Atomic radius is difficult to define – there is a small probability of finding an electron at an infinite distance from the nucleus.
- Often atomic radius is defined in terms of internuclear distance in a compound.

Methods of representing Size

covalent radii:

- half of the internuclear distance in a homonuclear X-X single bond.

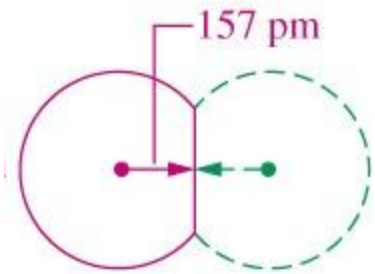
ionic radii:

- The ionic radius, r_{ion}, is the size of an ion (due to changes in effective nuclear charge; for cations, $r_{ion} < r_{cov}$; whilst for anions, $r_{ion} > r_{cov}$). These values are estimated from electron density maps determined from crystallographic data.

van der Waals Radii:

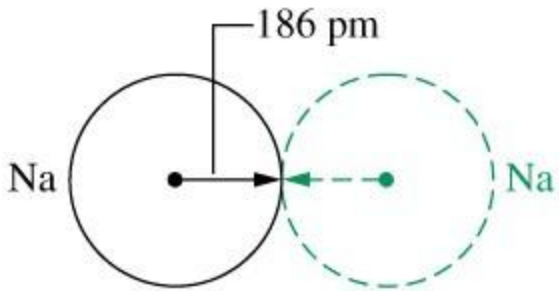
- half of the distance of closest approach of two non-bonded atoms. The forces holding the atoms together are weak London Dispersion forces(more to come…).

Covalent radius:



Metallic radius:

(metals often don't form covalent bonds)



Ionic radius:

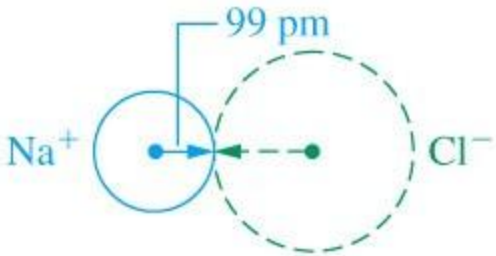


Fig. 9-3
Pg. 347

Periodic Trends in Atomic Radii

In order to understand trends in atomic radii, we must consider the shielding or screening effect of the core electrons.

<u>Effective nuclear charge</u>:

$$Z_{eff} = Z - S$$

- Z = atomic number
- S = number of inner electrons that *screen* an outer electron

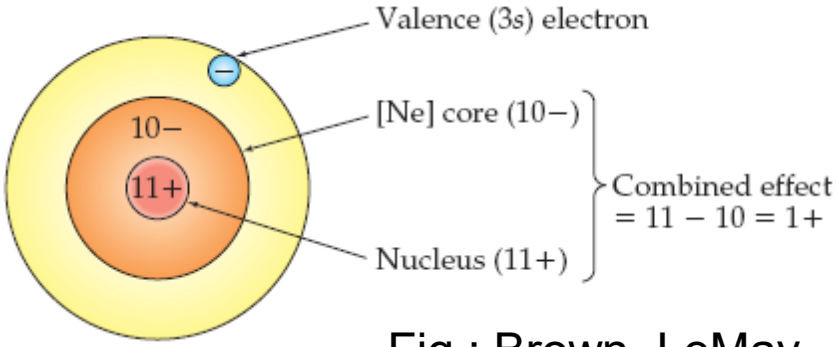


Fig.: Brown, LeMay

1. **Variation of atomic radii within a group of the periodic table.**

The more electronic shells in an atom, the larger the atom. Atomic radius increases from top to bottom in a group.

2. **Variation of atomic radii within a period of the periodic table.**

The atomic radius decreases from left to right through a period of elements.

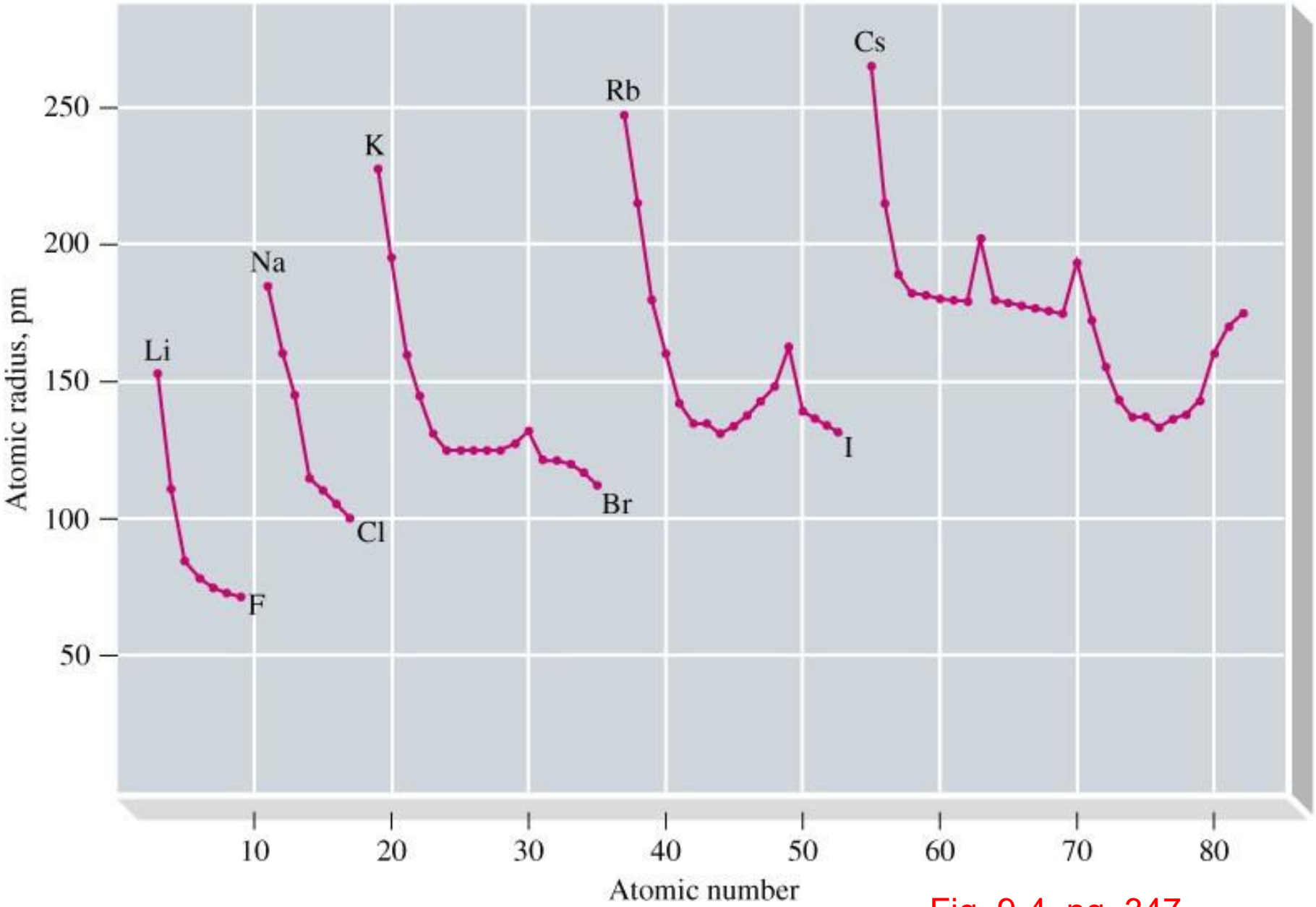

Atomic radius, pm
250
200
150
100
50
Li
F
Na
Cl
K
Br
Rb
I
Cs
10
20
30
40
50
60
70
80
Atomic number

Fig. 9-4, pg. 347

Ionic Radius

- Cations are smaller than the atoms from which they are formed.

Isoelectronic cations have the same number of electrons in identical configurations.

Na^+ and Mg^{2+} both have closed shells

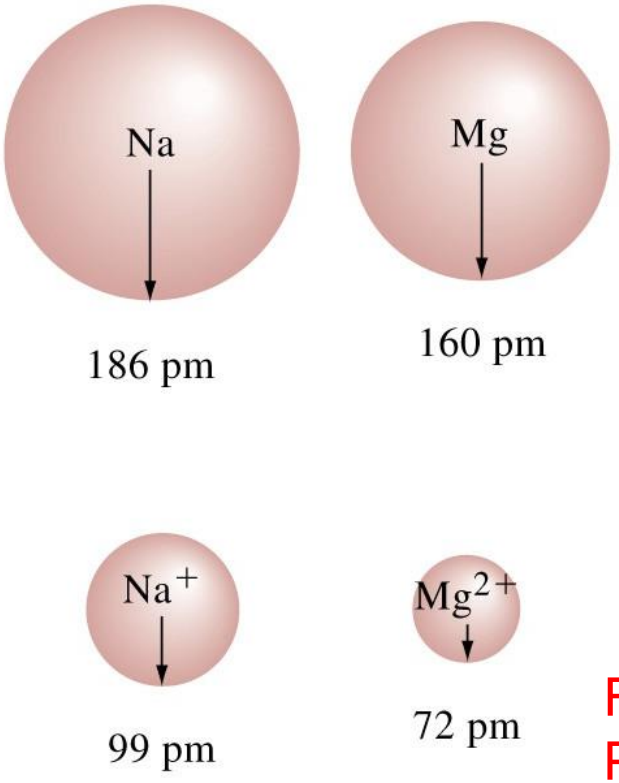


Fig. 9-6
Pg. 351

For isoelectronic cations, the more positive the ionic charge, the smaller the ionic radius.

- Anions are larger than the atoms from which they are formed.

- For isoelectronic anions, the more negative the charge, the larger the ionic radius.

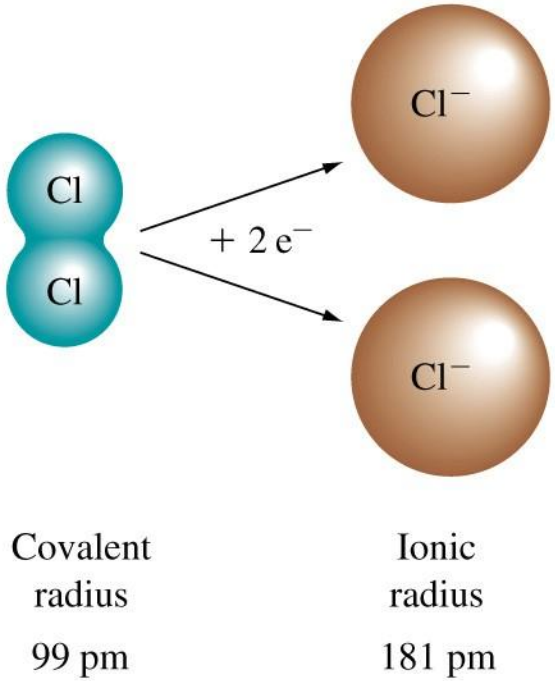
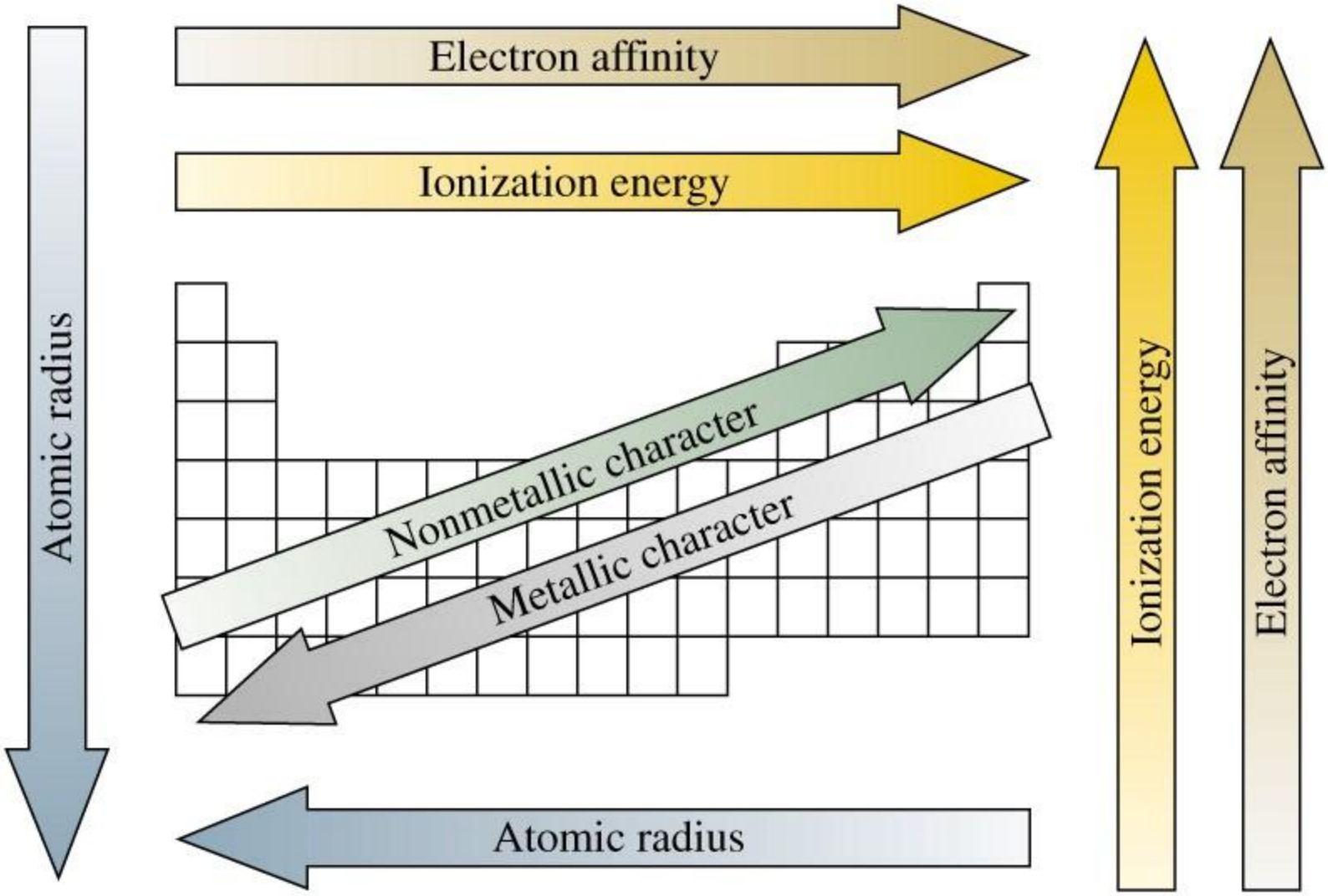


Fig. 9-7
Pg. 351

Summary



Source: Petrucci

Class Demonstrations: Reaction of Mg, Ca, Na and K with water

Although not a gas phase process, the trend in ionization energy is related to the trend in reactivity of the active metals since it involves the metals losing electrons:

$$Na(g) + I_1 \rightarrow Na^+(g) + e^- \ (I_1 \text{ first ionization energy})$$

$$Na(s) + H_2O(l) \rightarrow Na^+(aq) + OH^-(aq) + 0.5H_2(g)$$

Reactivity of active metals with water at room temperature

Na	**Mg**
K	Ca

Class exercise

Arrange the following atoms in order of increasing ionization energy:

Si, F, Na, O

Class exercise

Arrange the following atoms in order of increasing ionization energy:

Si, F, Na, O

Na<Si<O<F

Class exercise

Arrange the following atoms in order of increasing electron affinity

Cl, As, P, Ba

Class exercise

Arrange the following atoms in order of increasing electron affinity

Cl, As, P, Ba

Ba<As<P<Cl

Class exercise

Arrange the following atoms in order of increasing atomic radius:

Mg, N, Rb, Be

Class exercise

Arrange the following atoms in order of increasing atomic radius:

Mg, N, Rb, Be

N<Be<Mg<Rb

Class exercise

Arrange the following ions in order of increasing ionic radius:

Al^{3+}, Cl^{-}, Na^{+}, F^{-}

Class exercise

Arrange the following ions in order of increasing ionic radius:

Al^{3+}, Cl^{-}, Na^{+}, F^{-}

Al^{3+} < Na^{+} < F^{-} < Cl^{-}

10 e^{-}, 10 e^{-}, 10 e^{-}, 18 e^{-}

13 p^{+}, 11 p^{+}, 9 p^{+} 17 p^{+}

Ionic bond

Metals have low ionization energies and tend to lose electrons to form positive ions

Main group metals lose all their **valence electrons** to form cations:

If this occurs during a reaction it is an **oxidation.**

$Na^{\bullet} \longrightarrow Na^{+} + e^{-}$ sodium

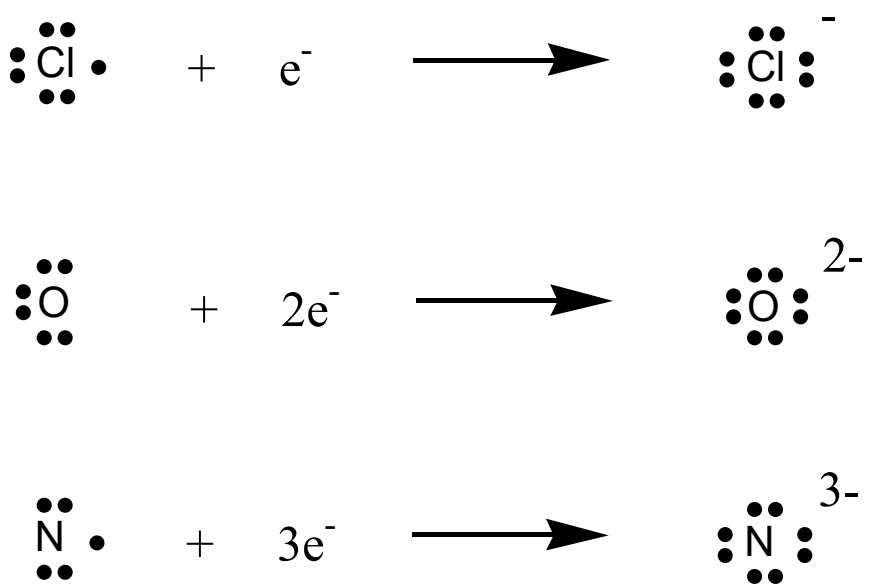
$Mg: \longrightarrow Mg^{2+} + 2\ e^{-}$ magnesium

$:Al^{\bullet} \longrightarrow Al^{3+} + 3\ e^{-}$ aluminum

Non-metals have high ionization energies and (except for the noble gases) tend to acquire electrons

Non-metals gain electrons to obtain the noble gas configuration and form anions:

The gain of electrons in a reaction is called **reduction.**



Ionic Compounds

- **Octet Rule:**
- When ionic bonds are formed, metals lose valence electrons so that the electron configuration becomes that of the nearest noble gas.
- Non-metals gain electrons to achive the noble gas electron configuration (8 or 2 valence electrons)

Note: There are many exceptions to the octet rule, but, it remains useful for introducing many important concepts of bonding.

Ions - Metals versus Nonmetals

1	2										13	14	15	16	17	18
H^+															H^-	NOBLE GASES
Li^+													N^{3-}	O^{2-}	F^-	
Na^+	Mg^{2+}	Transition metals									Al^{3+}		P^{3-}	S^{2-}	Cl^-	
K^+	Ca^{2+}				Cr^{3+}	Mn^{2+}	Fe^{2+} Fe^{3+}	Co^{2+}	Ni^{2+}	Cu^+ Cu^{2+}	Zn^{2+}			Se^{2-}	Br^-	
Rb^+	Sr^{2+}									Ag^+	Cd^{2+}		Sn^{2+}		Te^{2-}	I^-
Cs^+	Ba^{2+}								Pt^{2+}	Au^+ Au^{3+}	Hg_2^{2+} Hg^{2+}		Pb^{2+}	Bi^{3+}		

Brown, LeMay

Class Exercise

Write balanced equations for the reaction of:

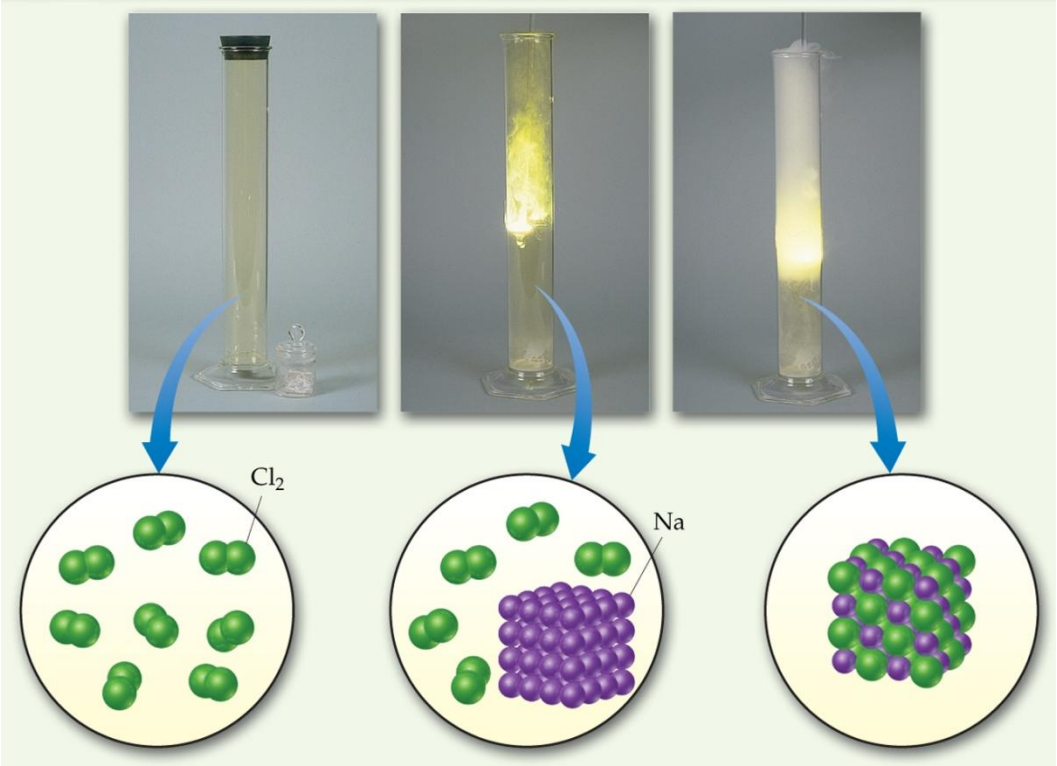
- sodium with bromine
- magnesium with chlorine
- aluminum with sulfur
- strontium with fluorine
- hydrogen with calcium
- nitrogen with lithium

Energetics of Ionic Bonding

As we have seen it takes 495 kJ/mol to remove electrons from sodium (first ionization energy of sodium).

We get 349 kJ/mol back by giving the electrons to chlorine (remember the exothermic electron gain).

Energetics of Ionic Bonding

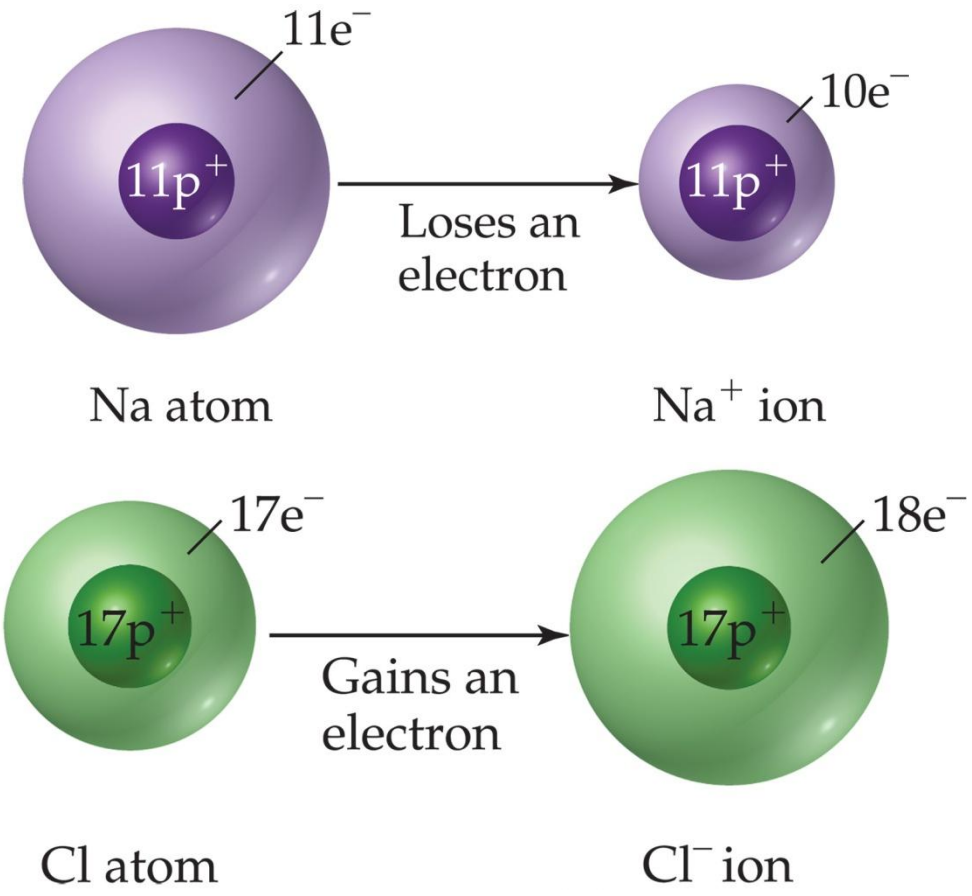


Brown; LeMay, Figure 7.2

- But these numbers don't explain why the reaction of sodium metal and chlorine gas to form sodium chloride is so exothermic!

Energetics of Ionic Bonding

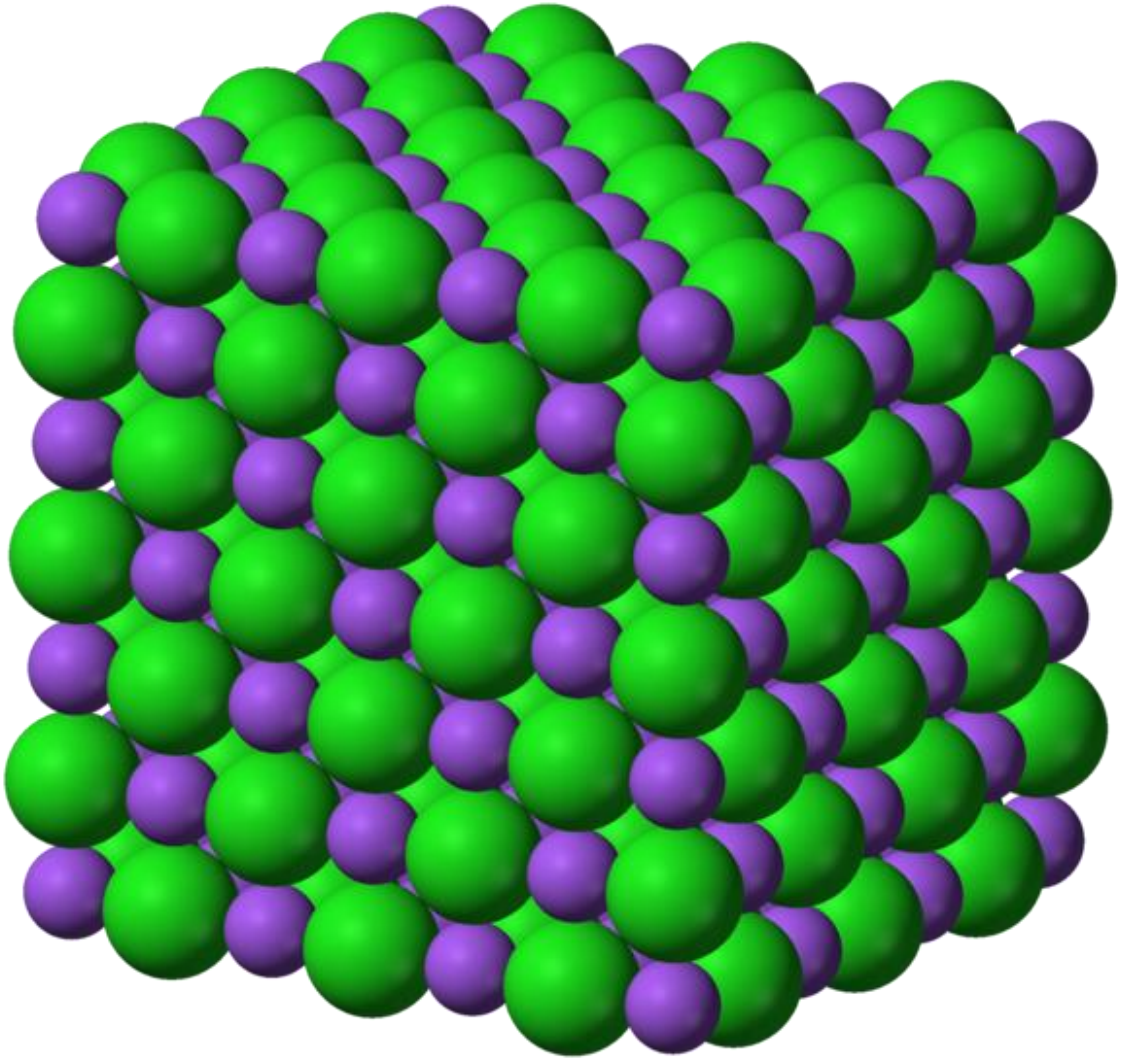
What is as yet unaccounted for is the electrostatic attraction between the newly formed sodium cation and chloride anion.



Energetics of Ionic Bonding

- Negative anions are attracted to positive cations.
- The result is an **ionic bond.**
- A three-dimensional **crystal lattice** of anions and cations is formed.
- The energy released upon forming the ionic crystal from isolated gas phase ions is known as **lattice energy**

Space filling View of NaCl

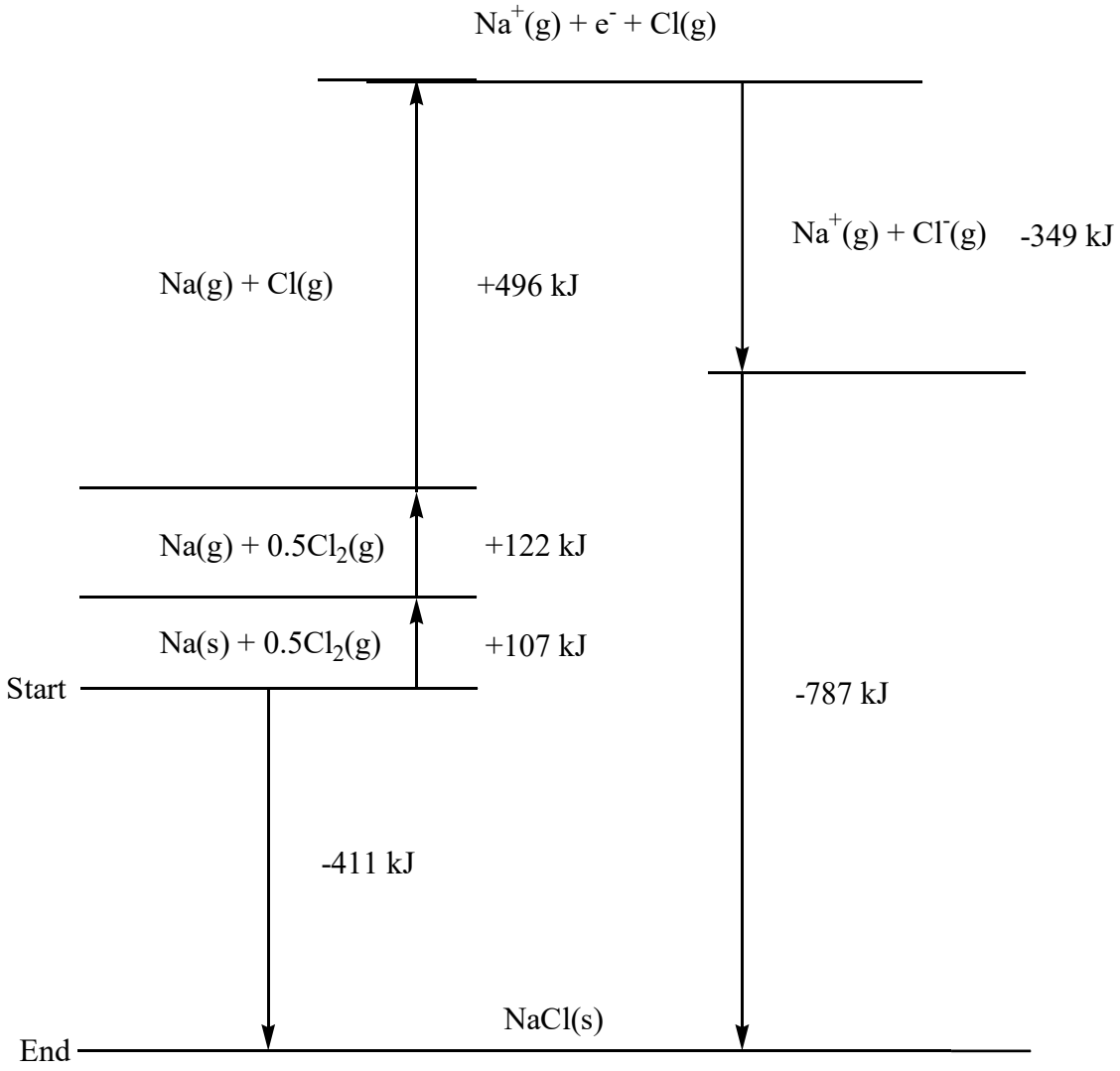


Energetics of Ionic Bonding

Lattice energies of the crystalline solids are generally very high.

As a result ionic compound tend to be hard (ceramics) and high melting solids…

e.g. would aluminum oxide (or sodium chloride) be used as a lubricant or rather as an abrasive?



Determining the Lattice Energy using the Born-Haber cycle

Structure- Properties Relationship

Class exercise: Rationalize the trends in lattice energies, melting and boiling points as well as aqueous solubilities.

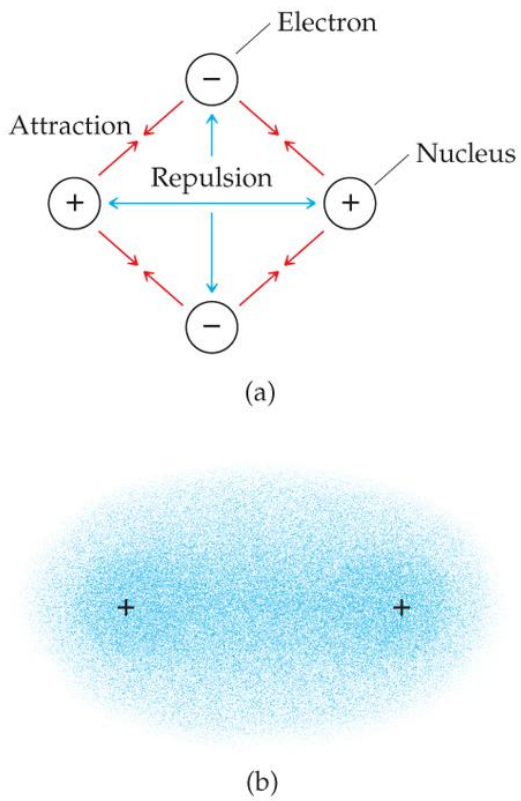
Compound	Lattice energy (kJ/mol)	Melting point (°C)	Boiling Point (°C)	Solubility (g/100mL H_2O)
LiF	1,049	848	1673	0.13
NaCl	787	801	1465	36
KBr	691	734	1435	67
RbI	632	656	1300	165

Compound	Lattice energy (kJ/mol)	Melting point (°C)	Boiling Point (°C)	Solubility (g/100mL H_2O)
Na_2O	2481	1134	decomposition	decomposition
MgO	3795	2825	3600	insoluble
Al_2O_3	15916	2054	2977	insoluble
ZrN	7633	2952	_____	insoluble

Formation of Molecular Compounds - Sharing of Electrons

- For some nonmetals it is not feasible to gain the number of electrons required to reach an octet.
- E.g. Carbon would have to gain 4 electrons!!!
- Carbon (and many nonmetals) share valence electrons so that they are surrounded by 8 electrons (hydrogen by 2 electrons)
- This is called **covalent bonding**

Covalent Bonding in Dihydrogen

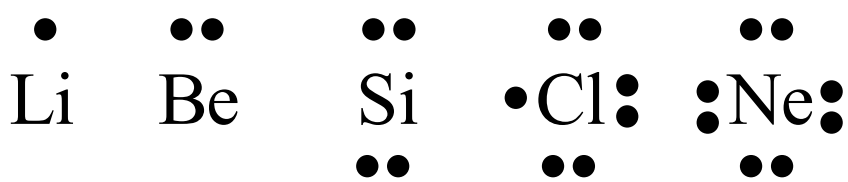


Brown; LeMay Figure 7.4

- In these bonds atoms share electrons.
- There are several electrostatic interactions in these bonds:
  - Attractions between electrons and nuclei
  - Repulsions between electrons
  - Repulsions between nuclei

Electron Dot Formulas (Lewis structures)

- It is often useful to represent the **valence electrons** (electrons in the outermost shell) as dots around the symbol for the elements (Important for chemical bonding!!!)
- These are called electron dot structures or Lewis structures (Lewis diagrams)



Lewis dot structures for atoms are straight forward!

Class Exercise

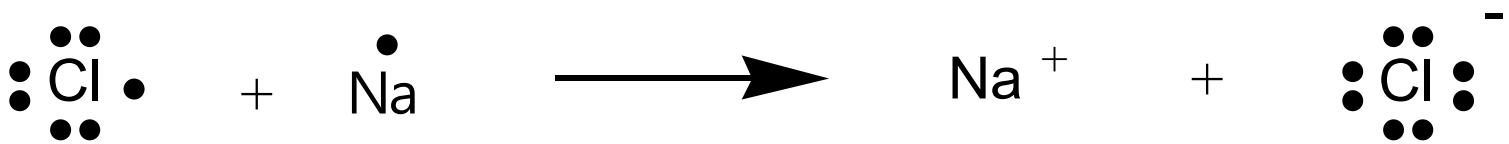
- How many valence electrons for each atom:
  - Potassium
  - Selenium
  - Iodine
  - Barium

Class Exercise

- Write electron dot formulas for each:
  - Calcium
  - Boron
  - Bromine
  - Sulfur

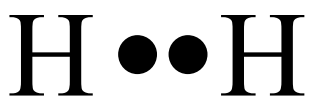
Chemical Bonding and Lewis structures

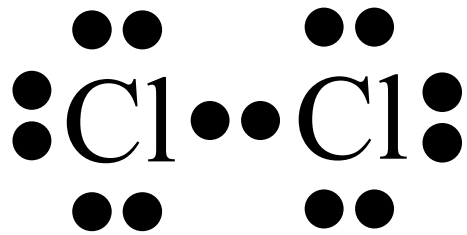
- Ionic bond: complete electron transfer between atoms yields ions with stable octets (or duets), Electrostatic attraction is the main force.



Chemical Bonding and Lewis structures

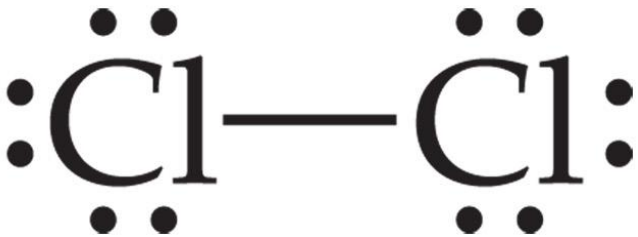
- Covalent bond: sharing of electrons so each atom obtains eight electrons (or two for hydrogen)





A **bond** is represented as a **solid line** between atoms. It represents a bonding electron pair.
A non-bonding electron pair is represented by a pair of dots.

Lewis Structures of Molecules



Lewis structures are representations of molecules showing all valence electrons - bonding and nonbonding.

Lewis Structures of Molecules Part 1 (simple molecules)

- Draw each atom along with it's valence electrons
- Make one Bond between the central atom and each terminal atom by pairing up electrons
- Pair electrons to make double bonds if necessarily to complete octet for all atoms (maximum of four bonds per atom)

Lewis structures of Molecules

Learn by doing!!!

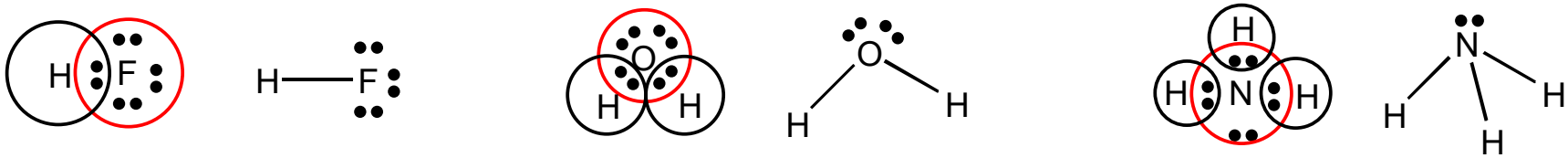
Write Lewis structures for the following molecules:

HF

H_2O

NH_3

Answers:



A bond is represented as a solid line between atoms.

Lewis structures of Molecules

Learn by doing!!!

Write Lewis structures for the following molecules:

N_2

CO_2

$[NH_4]^+$